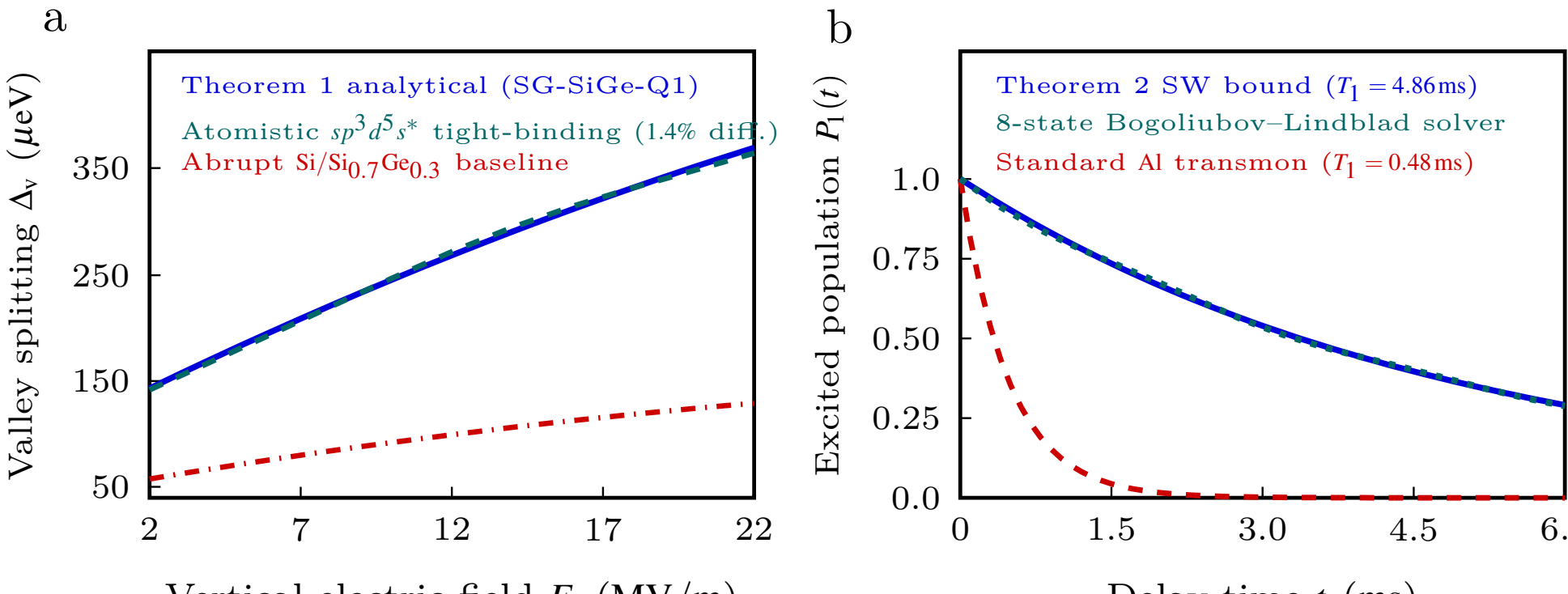
a
Theorem 1 analytical (SG-SiGe-Q1)
Atomistic $sp^3d^5s^*$ tight-binding (1.4% diff.)
Abrupt $Si/Si_{0.7}Ge_{0.3}$ baseline
Valley splitting Δ_v (μeV)
350
250
150
50
2
7
12
17
22
b
Theorem 2 SW bound ($T_1 = 4.86$ms)
8-state Bogoliubov–Lindblad solver
Standard Al transmon ($T_1 = 0.48$ms)
Excited population $P_1(t)$
1.0
0.75
0.50
0.25
0.0
0
1.5
3.0
4.5
6.